

MATHEMATICS BY D.K. SINGH

JEE MOCK TEST 2

Time: 1 Hour 15 minutes

Maximum Marks: 100

Read the instructions carefully before answering:

1. This paper contains two sections: **Section 1** and **Section 2**.
2. **Section 1** consists of 20 Single Correct Choice Questions (Questions 1 to 20).
3. **Section 2** consists of 5 Integral/Numerical Answer Type Questions (Questions 21 to 25).
4. Filling up the responses carefully on the OMR sheet is mandatory.

Section 1: Single Correct Choice Type (Questions 1 to 20)

Marking Scheme:

- **Full Marks:** +4 if only the correct option is chosen.
- **Zero Marks:** 0 if none of the options are chosen.
- **Negative Marks:** -1 in all other cases.

1. In a triangle ABC , if $A - B = 120^\circ$ and $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{1}{32}$, then $\cos C$ equals:
(A) $3/4$ (C) $5/6$
(B) $2/3$ (D) $7/8$
2. The value of the expression $\frac{1}{\cos 290^\circ} + \frac{1}{\sqrt{3} \sin 250^\circ}$ is:
(A) $\frac{\sqrt{3}}{4}$ (C) $\frac{2}{\sqrt{3}}$
(B) $\frac{4}{\sqrt{3}}$ (D) $\frac{\sqrt{3}}{2}$
3. If $a \cos^3 \alpha + 3a \cos \alpha \sin^2 \alpha = m$ and $a \sin^3 \alpha + 3a \cos^2 \alpha \sin \alpha = n$, then $(m+n)^{2/3} + (m-n)^{2/3}$ is equal to:
(A) $2a^2$ (C) $2a^{2/3}$
(B) $2a^{1/3}$ (D) $2a^3$
4. If $x = \sin \alpha$, $y = \sin \beta$, $z = \sin(\alpha + \beta)$, then $\cos(\alpha + \beta) =$
(A) $\frac{x^2+y^2+z^2}{2xy}$ (C) $\frac{z^2-x^2-y^2}{2xy}$
(B) $\frac{x^2+y^2-z^2}{xy}$ (D) $\frac{z^2-x^2-y^2}{xy}$

5. The expression $\frac{\cos 6x + 6 \cos 4x + 15 \cos 2x + 10}{\cos 5x + 5 \cos 3x + 10 \cos x}$ is equal to:

- (A) $\cos 2x$ (C) $\cos^2 x$
 (B) $2 \cos x$ (D) $1 + \cos x$

6. $\cos(A + B) + \sin(A - B) = ?$

- (A) $(\cos A - \sin A)(\cos B + \sin B)$ (C) $(\cos A - \sin A)(\cos B - \cos B)$
 (B) $(\sin A - \cos A)(\cos B + \sin B)$ (D) $(\cos A + \sin A)(\cos B - \sin B)$

7. Let $\frac{\sin 3\theta}{\cos 2\theta} = p$ where $\theta \in \left(\frac{18\pi}{48}, \frac{23\pi}{48}\right)$ and $\frac{\sin 3\beta}{\cos 2\beta} = q$ where $\beta \in \left(\frac{13\pi}{48}, \frac{14\pi}{48}\right)$. Then:

- (A) $p > 0$ and $q > 0$ (C) $p < 0$ and $q < 0$
 (B) $p > 0$ and $q < 0$ (D) $p < 0$ and $q > 0$

8. Each of the four statements given below are either True or False.

I. $\sin 765^\circ = -1/\sqrt{2}$

II. $\csc(-1410^\circ) = 2$

III. $\tan \frac{13\pi}{3} = 1/\sqrt{3}$

IV. $\cot\left(-\frac{15\pi}{4}\right) = -1$

Indicate the correct order of sequence, where 'T' stands for true and 'F' stands for false:

- (A) FTFT (C) TFFF
 (B) FFTT (D) FTFF

9. Find the smallest natural 'n' such that $\tan(107n)^\circ = \frac{\cos 96^\circ + \sin 96^\circ}{\cos 96^\circ - \sin 96^\circ}$.

- (A) $n = 2$ (C) $n = 4$
 (B) $n = 3$ (D) $n = 5$

10. The value of $\frac{\sin \frac{\pi}{48} + \sin \frac{3\pi}{48} + \sin \frac{5\pi}{48} + \sin \frac{7\pi}{48}}{\cos \frac{\pi}{48} + \cos \frac{3\pi}{48} + \cos \frac{5\pi}{48} + \cos \frac{7\pi}{48}}$ is equal to:

- (A) $2 - \sqrt{3}$ (C) $\sqrt{2} - 1$
 (B) $2 + \sqrt{3}$ (D) $\sqrt{2} + 1$

11. Let $x = 2^{(\log_2 3)(\log_3 4) \dots (\log_{99} 100)}$, then the value of $\cos\left(\frac{x\pi}{4}\right) + \sin\left(\frac{x\pi}{4}\right)$ is equal to:

- (A) 1 (C) $\sqrt{2}$
 (B) $\frac{1}{\sqrt{2}}$ (D) -1

12. Number of solution(s) of the equation $\log_{\sqrt{x}}(x + |x - 3|) = \log_x(2x + 5 + 2|x - 3|)$ is:

- (A) 0 (C) 2
 (B) 1 (D) Infinitely many

13. Given $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$. Using this information or otherwise, choose the correct statements:

- (A) $8^{150} > 9^{100}$
 (B) $16^{75} > 81^{50}$
 (C) $\log_{2/3} 3 > 0$
 (D) None of these is correct

14. In the inequality below the value of the angle is expressed in radian measure. Which one of the inequalities below is true?

- (A) $\sin 1 < \sin 2 < \sin 3$ (C) $\sin 2 < \sin 1 < \sin 3$
 (B) $\sin 3 < \sin 2 < \sin 1$ (D) $\sin 3 < \sin 1 < \sin 2$
15. If θ is eliminated from the equations $x = a \cos(\theta - \alpha)$ and $y = b \cos(\theta - \beta)$, then $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \cos(\alpha - \beta)$ is equal to:
 (A) $\sec^2(\alpha - \beta)$ (C) $\cos^2(\alpha - \beta)$
 (B) $\csc^2(\alpha - \beta)$ (D) $\sin^2(\alpha - \beta)$
16. The number of real values of x satisfying the equation $3|x - 2| + |1 - 5x| + 4|3x + 1| = 13$ is:
 (A) 1 (C) 2
 (B) 4 (D) 3
17. The product of all values of x satisfying the equation $|x - 1|^{\log_3 x^2 - 2 \log_x 9} = (x - 1)^7$, is:
 (A) 162 (C) $\frac{81}{\sqrt{3}}$
 (B) $\frac{162}{\sqrt{3}}$ (D) 81
18. If $\tan \gamma = \sec \alpha \sec \beta + \tan \alpha \tan \beta$, then the greatest value of $\cos 2\gamma$ is:
 (A) -1 (C) $-\frac{1}{2}$
 (B) $\frac{1}{2}$ (D) 0
19. The complete set of values of x satisfying the inequality $\log_3 \left(\frac{|x^2 - 4x| + 3}{x^2 + |x - 5|} \right) \geq 0$ is:
 (A) $(-\infty, -\frac{2}{3}] \cup [\frac{1}{2}, 2]$
 (B) $(-\infty, -1] \cup [\frac{1}{2}, 2]$
 (C) $[\frac{1}{2}, \frac{2}{3}] \cup [1, \infty)$
 (D) $[-\infty, -\frac{1}{2}] \cup [\frac{2}{3}, 2]$
20. Let $f(x) = \sqrt{1 + x \sqrt{1 + (x + 1) \sqrt{1 + (x + 2)(x + 4)}}$ for $x > -1$, then the value of $\frac{f(\tan 22^\circ) f(\tan 23^\circ)}{\left| \log_{\frac{1}{5}} 12 + \left| \log_{\frac{1}{5}} 3 \right| - \left| \log_{\frac{1}{5}} 20 \right| \right|}$ is equal to:
 (A) $\frac{1}{\log_5 2}$ (C) 1
 (B) $\frac{1}{\log_5 4}$ (D) 2

Section 2: Integral/Numerical Answer Type (Questions 21 to 25)

Marking Scheme:

- **Full Marks:** +4 if only the correct integer/numerical value is written.
- **Zero Marks:** 0 if the question is unanswered.
- **Negative Marks:** -1 in all other cases.

21. If $\log_{10} \left(\frac{1025}{1024} \right) = a$ and $\log_{10}(2) = b$, then $\log_{10}(4100) = ma + nb$, where $m, n \in \mathbb{N}$. Find the value of $m + n$.

22. Find the sum of all integral values of x satisfying the inequality:

$$2 \log_{\frac{1}{4}}(x + 5) > \frac{9}{4} \log_{\frac{1}{3\sqrt{3}}}(9) + \log_{\sqrt{x+5}}(2)$$

23. Let $P(k) = \left(1 + \cos \frac{\pi}{4k}\right) \left(1 + \cos \frac{(2k-1)\pi}{4k}\right) \left(1 + \cos \frac{(2k+1)\pi}{4k}\right) \left(1 + \cos \frac{(4k-1)\pi}{4k}\right)$. If $P(5) + P(6) = \frac{7-\sqrt{a}-\sqrt{5}}{b}$ where $a, b \in \mathbb{N}$, find the value of $a + b$.

24. The value of $\frac{\sec 11^\circ \sec 19^\circ - 2 \cot 71^\circ}{\cot 79^\circ}$ is equal to:

25. The value of $\left(\frac{\tan \frac{\pi}{16} + \tan \frac{5\pi}{16} + \tan \frac{9\pi}{16} + \tan \frac{13\pi}{16}}{\sin \frac{\pi}{16} \cos \frac{\pi}{16} \left(\cos^4 \frac{\pi}{16} - \sin^4 \frac{\pi}{16} \right)} \right)^2$ is equal to:

MATHEMATICS BY D.K. SINGH

OMR RESPONSE SHEET

CANDIDATE DETAILS:

Candidate's Name: _____	Class/Batch: _____
Father's Name: _____	Gender: (M) (F) (O)
Roll Number: _____	Date of Test: _____

SECTION 1: SINGLE CORRECT CHOICE QUESTIONS (1 – 20)

- | | | | |
|--------------------|---------------------|---------------------|---------------------|
| 1. (A) (B) (C) (D) | 6. (A) (B) (C) (D) | 11. (A) (B) (C) (D) | 16. (A) (B) (C) (D) |
| 2. (A) (B) (C) (D) | 7. (A) (B) (C) (D) | 12. (A) (B) (C) (D) | 17. (A) (B) (C) (D) |
| 3. (A) (B) (C) (D) | 8. (A) (B) (C) (D) | 13. (A) (B) (C) (D) | 18. (A) (B) (C) (D) |
| 4. (A) (B) (C) (D) | 9. (A) (B) (C) (D) | 14. (A) (B) (C) (D) | 19. (A) (B) (C) (D) |
| 5. (A) (B) (C) (D) | 10. (A) (B) (C) (D) | 15. (A) (B) (C) (D) | 20. (A) (B) (C) (D) |

SECTION 2: INTEGRAL/NUMERICAL TYPE QUESTIONS (21 – 25)

Write the final calculated single integer/numerical value inside the designated answer box clearly.

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|-----------------------------------|-----------------------------------|-----------------------------------|
| Q.21 Answer: <input type="text"/> | Q.22 Answer: <input type="text"/> | Q.23 Answer: <input type="text"/> |
| Q.24 Answer: <input type="text"/> | Q.25 Answer: <input type="text"/> | |

Signature of the Candidate

Signature of the Invigilator